

Name: _____



Programmer

Task A: Algorithms

Algorithm 1

Step 1: Take two positive numbers, A and B .

Step 2: Draw a table with two columns. Put A at the top of the left column and B at the top of the right column.

Step 3: In the next row, in the left-hand column write the number that is half of A (ignoring remainders) and in the right-hand column write the number that is double B .

Step 4: Repeat Step 3 until you have a 1 in the left-hand column.

Step 5: Delete any rows where the left-hand column has an even number.

Step 6: Find the sum of the numbers left in the right-hand column.

1) Carry out algorithm 1 with the following numbers:

a) $A = 8, B = 10$

b) $A = 29, B = 34$

c) $A = 13, B = 17$

d) $A = 66, B = 56$

2) What does algorithm 1 do?

3) Write an algorithm for making a drink of squash or a cup of hot chocolate (more challenging). Give your algorithm to a partner and let them try to 'break' your algorithm. Is there a way for them to misinterpret your instructions?

Task B: Logic

1) 'If it is raining, I will use an umbrella.'

What can you conclude if:

- a) It is raining.
- b) It is not raining.
- c) I am using an umbrella.
- d) I am not using an umbrella.

2) At Oak National Academy, it is logic day. On logic day, all pupils either always tell the truth or always lie.

You meet Jacob and Jun. Jacob says, 'Exactly one of us is telling the truth today.'

Is it possible to state whether Jacob is telling the truth or not?

3) At Oak National Academy, it is logic day. On logic day, all pupils either always tell the truth or always lie.

You meet Laura and Lucas. Laura says, 'I am lying today and Lucas is telling the truth today.'

What can you deduce, if anything, about Laura and Lucas?

4) At Oak National Academy, it is logic day. On logic day, all pupils either always tell the truth or always lie.

You meet Sam and Sofia.

Sam says, 'One of us is lying and one is telling the truth.'

Sofia says, 'We are both doing the same thing and I am lying.'

What can you deduce, if anything, about Sam and Sofia?